The Problem

Modern GPS and mobile systems can push coupon offers to drivers in real time, a coffee shop discount when you're nearby, and a bar deal when you're out with friends. But irrelevant offers are annoying and get dismissed, wasting the advertiser's budget and training users to ignore future offers.

This dataset was collected via Amazon Mechanical Turk surveys in which respondents were placed in simulated driving scenarios (with a specific destination, time, weather, passengers, etc.) and asked: "Would you use this coupon?" The core research question is: given everything we know about the driver and the context, can we predict whether they will accept a coupon?

This is a binary classification problem; the answer is always Yes (Y=1) or No (Y=0).

Its Importance

Sending a coupon only when the driver is likely to accept it means higher redemption rates, lower wasted spending, and better customer experience. A model that predicts acceptance lets you target the right person, at the right moment, with the right offer.

Goal

Maximize coupon redemption rates by recommending offers only to drivers who are likely to accept them, reducing wasted advertising expenditure.

Exploratory Data Analysis

The dataset captures real-world driving scenarios in which participants were asked whether they would accept a coupon for a nearby venue, such as a coffee house, restaurant, or bar while driving. The target variable Y indicates whether the coupon was accepted (Y=1) or rejected (Y=0).

The goal of this EDA is to uncover patterns, distributions, and relationships that help explain what drives coupon acceptance, which will later inform predictive modelling. The analysis is structured into five phases:

- Data Structure and Overview
- Descriptive Statistics
- Visualization of Key Variables
- Correlation Analysis
- Univariate Analysis
- Cross Tabulation Analysis
- Outlier Detection

Data Structure and Overview

The dataset was loaded using PROC IMPORT in SAS and inspected using PROC CONTENTS.

The 26 variables fall into two categories: numeric and character (categorical). They can also be logically grouped by the kind of information they capture:

Contextual / Situational Variables

These describe the conditions when the coupon was offered:

- destination: where the driver is going (Home, Work, No Urgent Place)
- passenger: who is in the car (Alone, Friend(s), Kid(s), Partner)
- weather: weather conditions (Sunny, Rainy, Snowy)
- time: time of day (7AM, 10AM, 2PM, 6PM, 10PM)
- temperature: ambient temperature in Fahrenheit (30, 55, or 80)
- expiration: coupon validity window (2h = 2 hours, 1d = 1 day)
- coupon: type of coupon (Bar, Coffee House, Carry out & Take away, Restaurant(<20), Restaurant(20-50))

Demographic Variables

These describe who the driver is:

- gender, age, maritalStatus, has_children, education, occupation, income, car

Behavioral / Frequency Variables

These capture how often the driver visits similar venues:

- Bar, CoffeeHouse, CarryAway, RestaurantLessThan20, Restaurant20To50

Distance and Direction Variables

- direction_same: whether the coupon destination is in the same direction as the driver's destination (binary)
- direction_opp: opposite direction (binary, complement of direction_same)
- toCoupon_GEQ5min, toCoupon_GEQ15min, toCoupon_GEQ25min — binary flags indicating travel time to coupon destination

Target Variable

- Y: binary outcome: 1 = coupon accepted, 0 = coupon rejected

Descriptive Statistics

Numeric Variables Summary

PROC MEANS was used to produce descriptive statistics for all numeric variables. The results are summarized below.

The MEANS Procedure

Variable	N	N Miss	Mean	Median	Std Dev	Minimum	Maximum
temperature	12684	0	63.3017975	80.0000000	19.1544858	30.0000000	80.0000000
age	10349	2335	29.8878152	26.0000000	7.6972751	21.0000000	46.0000000
has_children	12684	0	0.4141438	0	0.4925930	0	1.0000000
toCoupon_GEQ5min	12684	0	1.0000000	1.0000000	0	1.0000000	1.0000000
toCoupon_GEQ15min	12684	0	0.5614948	1.0000000	0.4962235	0	1.0000000
toCoupon_GEQ25min	12684	0	0.1191265	0	0.3239500	0	1.0000000
direction_same	12684	0	0.2147588	0	0.4106711	0	1.0000000
direction_opp	12684	0	0.7852412	1.0000000	0.4106711	0	1.0000000
Y	12684	0	0.5684327	1.0000000	0.4953144	0	1.0000000

- Age has 2,335 missing values (18.41% of records), making it the only numeric variable with missing data. The mean age is approximately 30 years, with a median of 26, indicating a right skew toward younger drivers.

- toCoupon_GEQ5min has a constant value of 1 for all observations, meaning the coupon destination is always at least 5 minutes away. This variable carries no discriminatory information and can be dropped in modelling.

- Only 11.9% of observations involve a coupon destination 25+ minutes away (toCoupon_GEQ25min mean = 0.12), so most coupons are for nearby venues.

- The target variable Y has a mean of 0.57, meaning approximately 56.84% of coupons were accepted overall, a slight class imbalance favoring acceptance.

Categorical Variable Summary

PROC FREQ was used to produce frequency tables for all categorical variables. The key distributions are highlighted below.

Destination

destination	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Home	3237	25.52	3237	25.52
No Urgent Place	6283	49.53	9520	75.06
Work	3164	24.94	12684	100.00

Nearly half of all trips had no specific destination, which suggests these drivers are most susceptible to detours and may be more likely to accept coupons.

Coupon Type

coupon	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Bar	2017	15.90	2017	15.90
Carry out & Take away	2393	18.87	4410	34.77
Coffee House	3996	31.50	8406	66.27
Restaurant(20-50)	1492	11.76	9898	78.04
Restaurant(<20)	2786	21.96	12684	100.00

Coffee House coupons are the most frequently offered, accounting for nearly a third of all records. Upscale restaurant coupons (Restaurant 20-50) are the least common.

Weather

weather	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Rainy	1210	9.54	1210	9.54
Snowy	1405	11.08	2615	20.62
Sunny	10069	79.38	12684	100.00

Most observations (79%) occurred on sunny days, which naturally dominates the weather-related patterns seen throughout the analysis.

Notable data quality issue: The car variable has 12,576 blank values (99.15% of all records), with only 108 observations containing coded values (C, M, S, c, d). This variable is effectively unusable and should be dropped during cleaning.

Visualizations of key variables

Target Variable

Visualization 1: Coupon Acceptance Rate (Pie Chart)

Of the 12,610 total observations, 7,157 coupons were accepted (56.76%) and 5,453 were rejected (43.24%). The overall acceptance rate slightly favors acceptance, but the dataset is not severely imbalanced. This baseline rate serves as a reference point: any predictive model must outperform a naive classifier that always predicts acceptance.

Contextual / Situational Features

Visualization 2: Coupon Type vs. Acceptance Rate

Coupon type is one of the strongest predictors of acceptance. Carry out & Take away coupons had the highest acceptance rate (73.4%), followed closely by Restaurant(<20) at 70.8%. Coffee House coupons were accepted about half the time (49.9%). Bar coupons had a low acceptance rate of only 41.0%, and upscale Restaurant(20-50) coupons had the second-lowest rate at 44.2%. This pattern suggests that low-cost, convenient coupons are far more appealing to drivers than coupons requiring a financial or time commitment.

Visualization 3: Weather vs. Acceptance Rate

Weather has a clear and consistent effect on coupon acceptance. Sunny weather produced the highest acceptance rates and dominates the dataset (79% of observations). Rainy weather showed somewhat lower acceptance, and snowy weather had the lowest acceptance proportion among accepted coupons. Drivers appear less willing to detour to a venue in adverse weather conditions. However, this effect should be interpreted cautiously given the heavy imbalance in weather observations.

Visualization 4: Time of Day vs. Acceptance Rate

Time of day shows meaningful variation in acceptance rates. The 6PM time slot recorded the highest acceptance rate, followed by 10AM. The 2PM window had the lowest acceptance rate among acceptances. The 7AM and 10PM slots show roughly balanced acceptance vs. rejection. This may reflect the fact that drivers at 6PM are leaving work and are in a more relaxed, social mindset, making them more receptive to coupon offers.

Visualization 5: Temperature vs. Acceptance Rate

Acceptance rate increases with temperature. At 80°F, acceptances (approximately 31%) clearly outnumber rejections (approximately 20%), while at 30°F the gap narrows. This aligns with the intuition that warm weather encourages spontaneous stops for food or drinks. Since temperature is discrete (only 3 values), it acts more as a categorical proxy for season or weather comfort than a true continuous variable.

Visualization 6: Destination vs. Acceptance Rate

Drivers with no urgent destination show a notably higher coupon acceptance rate (approximately 32% of all accepted coupons) compared to those heading home or to work (approximately 12-13% each). This is a strong and intuitive finding: drivers who are simply driving around are far more flexible about making a detour than those with a committed destination. This variable will likely be an important feature in the predictive model.

Visualization 7: Passenger Type vs. Acceptance Rate

The presence of friends in the car is strongly associated with coupon acceptance. Drivers accompanied by friends had the highest acceptance boost relative to rejections, followed by those travelling with a partner. Drivers travelling alone showed the largest proportion of rejections relative to their total count. Interestingly, drivers with kids in the car had the lowest acceptance rate overall, which may reflect the constraints of travelling with children.

Visualization 8: Coupon Expiration vs. Acceptance Rate

Coupons valid for one day (1d) were accepted at a substantially higher rate than those expiring in two hours (2h). The 1-day coupons show a clear acceptance majority, while the 2-hour coupons show a more balanced or even rejection-leaning split. This is logical: a longer window removes the urgency pressure, allowing drivers to plan a visit at their convenience rather than feeling forced to detour immediately.

Demographic Features

Visualization 9: Age Group vs. Acceptance Rate

Younger drivers (ages 21 and 26) show the highest acceptance rates relative to rejections and also represent the largest share of accepted coupons. As age increases, both the volume and the acceptance-to-rejection ratio decline. Drivers aged 46 represent the smallest and least receptive group. This pattern suggests that coupon marketing campaigns would be most effective when targeted at younger, more impulsive consumers.

Visualization 10: Income Level vs. Acceptance Rate

The income-acceptance relationship is not strictly monotonic. Low-to-middle income brackets ($12,500–$49,999) tend to show higher acceptance rates, which makes sense as discount coupons provide more utility to budget-conscious consumers. The $25,000–$37,499 bracket recorded the highest acceptance proportion. Interestingly, the highest income bracket ($100,000+) also shows relatively high acceptance, potentially due to Coffee House or convenience preferences. The mid-to-upper income range ($62,500–$87,499) showed the lowest acceptance rates.

Distance and Direction Features

Visualization 11: Same Direction as Coupon vs. Acceptance

This is one of the most striking visualizations. When the coupon destination is NOT in the same direction as the driver's route (direction_same = 0, which represents 78.5% of observations), acceptances still outnumber rejections but by a smaller margin. When the coupon IS in the same direction (direction_same = 1), the acceptance rate spikes dramatically, accepted coupons nearly double the rejection count. This confirms the intuitive insight that drivers are far more willing to accept a coupon if it requires no significant detour.

Visualization 12: Coupon Destination >=25 Minutes Away vs. Acceptance

When the coupon destination is less than 25 minutes away (toCoupon_GEQ25min = 0, representing 88% of observations), acceptances dominate clearly. When the destination is 25+ minutes away, both acceptance and rejection rates drop (fewer such scenarios exist), but the rejection rate becomes more competitive. This confirms that proximity is a key driver of acceptance: the closer the venue, the more likely the driver will take the detour.

Correlation Analysis

PROC CORR was used to compute pairwise Pearson correlation coefficients among all numeric variables. Only statistically significant correlations ($p < 0.05$) are discussed.

Variable	*Correlation With Y*	*P-value*	*Interpretation*
temperature	0.05939	<.0001	Warmer weather slightly increases acceptance
age	-0.03005	0.0023	Older drivers are slightly less likely to accept
has_children	-0.04506	<.0001	Parents are slightly less likely to accept
toCoupon_GEQ15min	-0.08269	<.0001	Moderate distance reduces acceptance
toCoupon_GEQ25min	-0.10814	<.0001	Farther destinations reduce acceptance
direction_same	0.01493	0.0936	Not significant at 5% level

All significant correlations with Y are weak in magnitude ($|r| < 0.11$), indicating that no single numeric variable is a strong linear predictor of acceptance on its own. This suggests that the predictive model will need to capture non-linear interactions and rely more heavily on categorical variables.

Notable Inter-variable Correlations

- direction_same and direction_opp are perfectly negatively correlated ($r = -1.000$), as expected since they are binary complements of each other. Only one should be used in modelling.

- toCoupon_GEQ15min and toCoupon_GEQ25min are moderately positively correlated ($r = 0.321$), which is expected since a 25-minute trip necessarily implies a 15-minute trip.

- age and has_children are moderately positively correlated ($r = 0.422$), reflecting that older drivers are more likely to have children.

- toCoupon_GEQ5min has undefined correlations (shown as missing) because it is a constant (all values = 1), confirming it should be excluded from all further analysis.

Univariate Analysis

PROC UNIVARIATE with the NORMAL option was applied to temperature and age to assess their distributional properties in detail.

Temperature

- Mean: 63.26°F | Median: 80°F | Mode: 80°F

- Skewness: -0.64 (left/negative skew), Kurtosis: -1.026 (platykurtic: flatter than normal)

- Normality tests all rejected (K-S $D = 0.322$, $p < 0.01$), confirming the distribution is non-normal, which is expected since the variable is discrete with only 3 values.

- The IQR is 25°F (Q1=55, Q3=80), with 80°F being both the median and 75th–100th percentile, meaning most observations occurred in warm weather.

Age

- Mean: 29.86 years | Median: 26 years | Mode: 21 years
- Skewness: +0.56 (right/positive skew), Kurtosis: -0.745 (slightly platykurtic)
- Normality tests all rejected (K-S D = 0.197, $p < 0.01$), indicating the age distribution is non-normal; it is right-skewed with a heavier tail toward older drivers.
- The IQR spans 15 years (Q1=21, Q3=36). The most extreme values are 21 (minimum, also the 25th percentile) and 46 (maximum), confirming the age range is narrow and discrete.
- 18.44% of age records are missing, which must be handled before modelling.

Cross Tabulation Analysis

PROC MEANS with CLASS statements was used to compute acceptance rates across variables, revealing interaction effects that single-variable visualizations cannot capture.

Acceptance Rate by Passenger Type and Coupon Type

The interaction between who is in the car and the type of coupon is particularly revealing:

- Friends + Bar: 55.8% acceptance. The highest bar acceptance rate across all passenger types, reflecting the social nature of bar visits.
- Partner + Restaurant(20-50): 63.1% acceptance. Couples are significantly more likely to accept upscale dining coupons.
- Kids + Bar: only 21.1% acceptance. The lowest acceptance rate in the entire matrix, confirming that parents are reluctant to take children to bars.
- Friends + Restaurant(<20): 80.1% acceptance. By far the highest acceptance rate overall, combining a social setting with an affordable and convenient venue.

Acceptance Rate by Expiration and Coupon Type

Expiration interacts differently with each coupon type:

- Restaurant(<20) with 1-day expiry: 83.1%. The highest acceptance rate in this table suggesting that affordable restaurants with relaxed timing are the most attractive.
- Restaurant(20-50) with 2-hour expiry: 29.9%. The lowest acceptance rate, showing that expensive restaurants with tight deadlines, are strongly rejected.
- Coffee House coupons benefit substantially from a longer expiry window: acceptance rises from 43.2% (2h) to 58.3% (1d).

Average Acceptance Rate by Weather x Time of Day

The interaction between weather and time of day produces notable patterns:

- Sunny + 2PM: 67.8%. One of the highest acceptance rate in the matrix. Warm, clear afternoons are the optimal conditions for coupon acceptance.
- Snowy + 10AM: 74.8%. Surprisingly high, possibly because morning commuters in cold weather welcome the idea of a warm coffee stop.
- Rainy + 10PM: only 37.1%. The worst combination, as late-night rainy conditions strongly discourage detours.

Outlier Detection

Outlier detection was performed using the IQR (Interquartile Range) method, which defines outliers as observations falling below Q1 - 1.5×IQR or above Q3 + 1.5×IQR. Two numeric variables were analyzed: temperature and age.

temp_outlier	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	12610	100.00	12610	100.00

No temperature outliers were detected. This is expected because temperature only takes three fixed discrete values (30, 55, 80°F), all of which fall comfortably within the IQR bounds. The box plot confirms symmetric distribution across acceptance groups with no extreme points.

age_outlier	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	10285	81.56	10285	81.56
1	2325	18.44	12610	100.00

The 2,325 records flagged as age outliers are entirely due to missing values in the age column. In SAS, missing numeric values are treated as the smallest possible value (effectively -infinity) when evaluating logical comparisons, causing them to fall below the lower IQR bound and be incorrectly classified as outliers. These are not true outliers; they are missing observations. The appropriate treatment is to impute or exclude these records in the data cleaning phase, not to discard them as anomalies.

The age box plot shows that the distribution is right-skewed, with rejected coupons (Y=0) showing a slightly higher median age than accepted ones (Y=1). Younger drivers tend to accept more coupons, consistent with the bar chart findings in Visualization 9.

Conclusion

This EDA has revealed a rich set of patterns that strongly predict whether a driver will accept a coupon. The most powerful individual predictors appear to be contextual factors, particularly coupon type, the direction of the venue relative to the driver's route, the driver's destination, passenger type, and the expiration window. Demographic factors such as age and income also play a role, but their effects are more nuanced and non-linear.

The dataset is largely clean for most variables, with the notable exceptions of the car variable (effectively unusable due to near-total missingness) and the age variable (18.44% missing). The toCoupon_GEQ5min variable should be dropped due to zero variance, and direction_opp should be dropped as it is redundant with direction_same.

These findings will directly inform the feature selection and engineering phase, where meaningful interaction terms (e.g., passenger × coupon type, expiration × destination) can be constructed to enhance the predictive model.

Data Cleaning

1. Introduction

This report presents a complete data preprocessing pipeline applied to the dataset WORK.coupons_nodup.

The goal is to assess data quality, handle missing values, clean categorical variables, and prepare the data for modeling through a proper train-test split.

2. Initial Data Quality Assessment

2.1 Missing Values (Numerical Variables)

A preliminary analysis was performed using PROC MEANS to evaluate missing values in numerical variables (age and temperature).

- N represents the number of non-missing observations.
- NMISS indicates the number of missing values.
- MIN and MAX provide a range of data.

This step helps identify the extent of missing data and detect potential data quality issues.

2.2 Missing Values Distribution (All Variables)

To analyze missing values across all variables:

- Custom formats were created using PROC FORMAT to classify values as:
    - **Missing**
    - **Not Missing**
- PROC FREQ was used to display the distribution of missing values across all columns.

This approach improves readability and provides a clear overview of missing data patterns in both numerical and categorical variables.

3. Data Cleaning

3.1 Dropping Irrelevant Variables

The variable car was removed due to an extremely high percentage of missing values, making it unreliable for analysis.

4. Train-Test Split

The dataset was split into training and testing sets using PROC SURVEYSELECT:

- Method: Simple Random Sampling (SRS)
- Split ratio: 80% training, 20% testing
- SEED=12345 ensures reproducibility

A new variable Selected was automatically generated:

- Selected = 1 → Training set
- Selected = 0 → Testing set

The split was verified using PROC FREQ to confirm the correct proportions.

5. Numerical Imputation

5.1 Mean Calculation

The mean of the variable age was computed using only the training dataset to avoid data leakage.

5.2 Imputation Process

Missing values in age were replaced with the computed mean:

- Applied to both training and testing datasets
- The same mean value from the training set was used for consistency

5.3 Rounding

The imputed values were rounded to the nearest integer to maintain realistic values for age.

6. Categorical Cleaning and Standardization

A macro %clean_subset was created to clean and standardize categorical variables efficiently.

6.1 Handling Missing Values

Missing categorical values were replaced with default values (e.g., "never", "less1") to ensure no empty entries remain.

6.2 Standardization

Categorical values were standardized using the following steps:

- Convert text to lowercase
- Remove extra spaces
- Map multiple representations into unified categories

Examples:

- ('never', 'nev', 'n') → **Never**
- ('less1', 'les', 'l') → **Less than once**
- ('1~3', '1') → **1 to 3 times**
- ('4~8', '4') → **4 to 8 times**
- ('gt8', 'g') → **More than 8 times**

This ensures consistency and improves model performance.

7. Final Validation

7.1 Missing Values Check

PROC MEANS was used again to confirm that missing values in the age variable were successfully handled.

7.2 Categorical Distribution

PROC FREQ was applied to:

- Verify the distribution of cleaned categorical variables
- Ensure no unexpected or inconsistent categories remain

8. Conclusion

This preprocessing pipeline successfully:

- Identified and analyzed missing data
- Removed irrelevant features
- Split the dataset into training and testing sets
- Applied proper numerical imputation without data leakage
- Cleaned and standardized categorical variables
- Validated the final dataset

The resulting datasets (train_final_clean and test_final_clean) are now clean, consistent, and ready for machine learning modeling.

Data Preparation & Feature Engineering

This outlines the final data preparation phases for the In-Vehicle Coupon Recommendation dataset. The primary goal of these steps was to transform raw data into a mathematically optimal format for a predictive model (such as Logistic Regression) while preventing data leakage. To ensure consistency, all feature engineering and encoding steps were encapsulated within a SAS Macro (%do_feature_engineering), allowing identical transformations to be applied independently to both the Training and Testing datasets.

1. Feature Engineering (Creating Business Insights)

Instead of relying solely on the raw variables, we engineered new features by combining existing columns. This helps the model capture complex relationships and real-world intuition.

- **Total_Distance_Level (Ordinal Scale):**
  - **Logic:** Summed the binary distance flags (toCoupon_GEQ5min, toCoupon_GEQ15min, toCoupon_GEQ25min).
  - **Intuition:** Consolidates multiple redundant flags into a single continuous/ordinal scale representing how far the destination is. A higher number indicates a further distance.

- **Coupon_Habit_Match (Binary Flag):**
    - **Logic:** Checks if the coupon type offered matches the driver's historical visiting habits. If the coupon is for a Coffee House, and the driver visits Coffee Houses 1 or more times a month, the flag is set to 1 (Match). Otherwise, 0.
    - **Intuition:** People are far more likely to accept a coupon for a place they already requent. This feature directly captures user affinity.
- **Companion_Logic (Binary Flag):**
    - **Logic:** Flags potentially illogical scenarios. Specifically, if the passenger is a Kid(s) and the coupon is for a Bar, the flag is set to 0 (Conflict). Otherwise, 1 (Logical).
    - **Intuition:** Captures the situational constraints. A driver is highly unlikely to accept a bar coupon if they have children in the car.
- **Bad_Weather_Home_Flag (Binary Flag):**
    - **Logic:** Set to 1 if the weather is 'Rainy' or 'Snowy' AND the driver's destination is 'Home'.
    - **Intuition:** If the weather is poor and a driver is already heading home, they are less likely to detour to use a coupon. This captures a specific high-rejection scenario.
- **Urgency_Score (Ordinal Scale 0-4):**
    - **Logic:** Combines the expiration time of the coupon with the distance to the venue.
    - **Intuition:** Evaluates the feasibility of using the coupon. A coupon that expires in 2 hours for a venue 25 minutes away creates high urgency (or low feasibility), whereas a 1-day expiration provides low urgency.

2. Categorical Encoding (Ordinal Encoding)

Machine learning models, particularly Logistic Regression, require numerical inputs. We transformed text-based frequency columns into mathematical scales.

- **Transformed Variables:** Bar, CoffeeHouse, CarryAway, RestaurantLessThan20, Restaurant20To50.
- **Mapping Logic:**
    - 'Never' → 0
    - 'Less than once' → 1
    - '1 to 3 times' → 2
    - '4 to 8 times' → 3
    - 'More than 8 times' → 4
- **Intuition:** This mapping preserves the inherent order (ordinality) of the data. The model can now understand that a "3" represents a higher frequency than a "1", allowing it to assign appropriate mathematical weights.

3. Dimensionality Reduction (Dropping Redundant Columns)

To prevent **Multicollinearity** (where independent variables are highly correlated with each other, which destabilizes Logistic Regression), the original columns used to create engineered features were dropped.

- **Dropped Variables:** toCoupon_GEQ5min, toCoupon_GEQ15min, toCoupon_GEQ25min. (These were perfectly summarized by the new Total_Distance_Level variable).

4. Numeric Scaling & Preventing Data Leakage

Features with large numeric ranges can disproportionately dominate the weights in algorithms like Logistic Regression. To fix this, we applied Standardization (Z-Score Scaling) to the continuous variables (temperature and age), giving them a Mean of 0 and a Standard Deviation of 1.

Crucial Methodological Step (Handling Train/Test Split): To ensure strict model evaluation validity, we prevented Data Leakage during scaling:

- **Training Data:** PROC STDIZE was run on the Train dataset. The calculated statistical parameters (Mean and STD) were extracted and saved to a separate file (WORK.scale_stats).
- **Testing Data:** Instead of recalculating the Mean and STD on the Test dataset (which would "leak" information about the test data distribution into the transformation), we used the METHOD=IN() option in SAS. This forced the scaler to transform the Test data using only the statistics learned from the Training data.

This ensures that the Test dataset remains a completely unseen, real-world simulation for final model evaluation.

Modeling

Logistic Regression Model

A **Logistic Regression model** was developed to predict whether a customer will **accept or reject a coupon** based on several behavioral and contextual features.

The model was trained using the training dataset and included both numerical and categorical variables such as customer demographics, environmental conditions, and engineered features like urgency and habit matching.

A stepwise feature selection method was applied to automatically choose the most relevant predictors, improving the model's efficiency and reducing unnecessary complexity.

Logistic Regression Evaluation

Probability Distribution Analysis

The predicted probabilities were analyzed to understand how confidently the model distinguishes between customers who accept and those who reject the coupon.

A good model shows a clear separation between the two groups, indicating strong predictive capability.

Confusion Matrix

The model's predictions were compared with actual outcomes using a confusion matrix.

This evaluation helps identify:

- Correct predictions (accept and reject)
- Incorrect predictions (false acceptance or rejection)

This provides a clear view of how well the model performs in real scenarios.

K-Nearest Neighbors (KNN Model)

A **K-Nearest Neighbors (KNN)** model was also implemented as an alternative approach.

This model classifies customers based on similarity to other customers, using distance between feature values. Unlike Logistic Regression, KNN does not assume a linear relationship between variables.

KNN Model Evaluation

Performance Metrics

The KNN model was evaluated using:

- **Sensitivity (Recall):** ability to correctly identify customers who accept the coupon
- **Specificity:** ability to correctly identify customers who reject the coupon

These metrics provide deeper insight into how well the model handles different classes.

Visual Analysis

A visual comparison between actual and predicted values was conducted to better understand the agreement between the model and real outcomes.

This helps in spotting patterns of misclassification.

Model Comparison

- **Logistic Regression**
  - Easier to interpret.
  - Works well when relationships are relatively linear.
  - Helps understand the impact of each feature.

- **KNN**
    - More flexible and it captures complex patterns.
    - Does not rely on assumptions about data distribution.
    - Can perform better with non-linear relationships.

Important Features

The Logistic Regression model identifies several key features that significantly influence coupon acceptance.

The most important feature is **Coupon_Habit_Match**, which has the strongest positive impact. This indicates that customers are far more likely to accept a coupon when it matches their habits.

Another influential feature is **Urgency_Score**, showing that customers are more likely to accept coupons when they feel time pressure.

Additionally, **Total_Distance_Level** has a negative effect, meaning that greater distance reduces the likelihood of acceptance.

These results suggest that **customer habits, urgency, and convenience** are the main drivers behind coupon acceptance decisions.

How does our solution help real-world decision making?

The developed model can support real-world decision making by helping businesses target the right customers with the right coupons at the right time.

By identifying key factors such as **customer habits**, **urgency**, and **distance**, companies can personalize their marketing strategies. For example, coupons can be sent to customers whose preferences match the offer, increasing the likelihood of acceptance.

Additionally, the model shows that customers are more responsive under time pressure, so businesses can use limited time offers to encourage faster decisions.

The negative impact of distance suggests that location-based targeting is important, meaning businesses should prioritize nearby customers to improve effectiveness.

Overall, this solution helps businesses **increase coupon acceptance rates, reduce wasted offers, and improve customer satisfaction** through more accurate and data-driven marketing decisions.